Enhancement of old images and documents by Digital Image Processing Techniques

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Abstract*—Documents can be a valuable source of information but often they suffer degradation problems, especially in the case of historical documents, such as strains, background of big variations and uneven illumination, ink seepage, etc. Binarization techniques should be applied to remove the noise and improve the quality of the documents. Collections of historical and old document images care commonly provided to public through digital libraries. Specialized processing is required to these document images for removing background noise in order to become more legible. A hybrid binarizatin approach is proposed in this paper for improving the quality for the old documents. Combination of global and local thresholding techniques are used for the same .Initially, a technique named global thresholding is applied to the whole image. The image area that still has background noise are detected and the technique is again re-applied to each area separately. Therefore, a better adaptability is achieved for the algorithm where various kinds of noise re exist in different areas of same image. Advantage of applying global thresholding, is that it avoids the computational and time cost of applying a local thresholding in the entire image. Hence it is indicated that this technique is pretty effective in removing background noise and improving the quality of degraded images.

***Index Terms*—Image enhancement, Iterative global thresholding, Local thresholding, hybrid binarization, Noise**

I. INTRODUCTION

There are libraries in the world which has collections of historical & ancient documents which are of great scientific & cultural importance. To maintain the quality of the originals it is essential that the documents are transformed into digital form,by doing this scholars are provided to have full access to the information. Degradation problems are quiet common in these documents. Few factors that impede (in many cases may disable) the legibility of the documents are strains, big variations, uneven illumination, presence of smear, seepage of ink, etc.

Before libraries expose them to public view it is important to remove noise from historical document images and improve their quality and appropriate filtering methods should be developed as well. Noise is considered anything that is irrelevant with the textual information (i.e., foreground) of the document image. . Binarization is used as a standard procedure to convert a grey-scale image to binary form in image analysis systems. An ideal binarization algorithm would be able to perfectly discriminate foreground from background, helping in removing any kind of noise that obstructs the legibility of the document image. Binary image is ideal for preprocessing like recognition of the contents by applying OCR techniques etc, discrimination of printed from handwritten text, etc. In the framework of a library collection of historical and ancient documents, the document images in many cases do not need further processing apart from removing the background noise and leave some “traces of time” behind, which is intended to be exposed to public view. In such cases its possible for the document images to remain in greyscale form.

For example consider the images of Figure 1.Figure 1a shows the original image, figure 1b result of the binarization procedure, and figure 1c the corresponding greyscale result after removing the background noise. The remaining noise and the text characters are smoothed a little bit in the latter case. While the characters are more legible, this has the consequence of making the background noise practically invisible. This option is supported by some binarization techniques.

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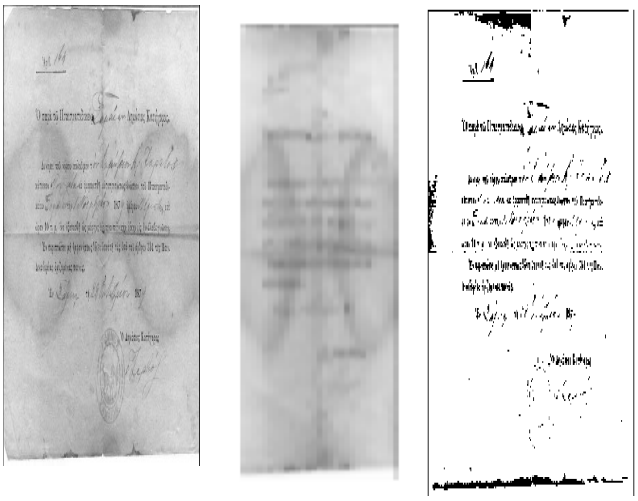


Figure 1.Removal of background noise from historical document images a)Original document b)Binarized document c)Result in grey scale form

Traditional binarization approaches can be divided into two main categories:

1. Global thresholding methods: According to global threshold, the pixels of the image are classified into background or text. Generally, such methods are simple and fast. If the background noise is unevenly distributed in the entire image, then they cannot be easily adapted. (e.g., smear or strains) [7-8].

2. Local thresholding methods: According to local threshold determined by their neighboring pixels, the pixels of the images are classified into background or text.

Adapting such methods are of more help and also can deal with different kind of noise existing in one image. On the other side its more time consuming and computationally expensive [9-10].

From another point of view, binarization approaches can be divided as follows:

1. General-purpose methods: Image can be dealt with this method. Hence, specific characteristics of document images are not taken into account.

2. Document image-specific methods: Advantage of document image characteristics (e.g., background pixels is the majority, foreground pixels are in similar grey-scale tones etc) are taken.

In many cases, such methods are variations of general-purpose approaches [9].

When dealing with historical document images, the latter approaches should be more effective. Recent results show that general-purpose methods can be more reliable under certain conditions [11].

Previously, we have presented an Iterative Global Thresholding (IGT) approach that is specifically designed for document images [6].

This method has the additional advantage of providing the option to maintain the image in grey-scale after the removal of background noise, apart from efficiency inherent in any global thresholding approach, a more familiar form for human readers.

A hybrid approach is proposed in this paper to combine the advantages of local and global thresholding. After the application of IGT to the document image, the areas that are more likely to still include significant amount of noise are selected and, then, IGT is re-applied to these areas separately. This is the main idea. Therefore, document images which have different kind of background noise, and which are unevenly distributed in the entire image can be processed more effectively. Additionally, in comparison to original local thresholding techniques, since only a limited number of areas (instead of the entire image) need to be processed separately, the time-cost of the approach remains on low level. In order to evaluate the proposed approach, a degree is represented in which the legibility of the image has been improved.

The rest of this paper is organized as follows: Section 2 includes related work and section 3 describes our approach in detail. Evaluation results are presented in section 4 while the main conclusions drawn from this study and future work directions are given in section 5.

II. LITERATURE S URVEY

There are many methods proposed in global and local thresholding. Based on the variance of pixel intensity, Otsu [7] proposed one of the earlier methods in image binarization. Bernsen [10] calculates local thresholds using neighbors.Standard deviation and local mean is used by Niblack [12]. A method is presented by Sauvola [9] which is specialized on document images that applies two algorithms in order to calculate a different threshold for each pixel. Leedham [8] compares some of the traditional methods on degraded document images, as far as the problem of historical documents is concerned, while a method using a combination of existing techniques is proposed by Gatos [13]. Shi [14] and Yan [15] is also applied to some historical documents from the US library of Congress. A serialization of k-means algorithm is implemented by Leydier [16] who works with colored document images. These above methods have also used per-processing or post- processing filters for improving the quality

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of the document image [13], except the basic binarization algorithm.

III. HYBRID BINARIZATION

A. The Proposed Approach

A combination of global and local thresholding. which is a hybrid approach for improving the quality of historical document images is proposed. A document image is applied a global thresholding approach (IGT) firstly. After that the areas which still contain noise are detected and re-processed separately. If we go more deep, the proposed algorithm consists of the following steps:

(i) IGT to the document image is applied.
(ii) Detect those areas in which we can still find background noise.
(iii) Then re-apply IGT to each detected area separately.

We avoid the cost of applying local thresholding to the entire image, by selecting only specific areas of the image for processing based on local thresholding.

B. Detection of Areas with Remaining Noise

A simple method is used for detection of areas that needs further processing. The areas that still contain background noise will include more black pixels on average in comparison with the other areas.

The image is divided into segments of fixed size nxn. The frequency of black pixel is calculated in each segment. The segments that satisfy the following criteria are, then, selected as:

$$f(S)>m+ks$$

where f(S) is the frequency of the black pixels in the segment S while m and s are the mean and the standard deviation of the black pixel frequency considering the segments of the entire page, respectively. The selected segments form areas by connecting neighboring segments in respect to their original position in the image. The row-by-row labeling algorithm is used [17].

C. Local Thresholding on Selected Areas

Based on local thresholding, the areas detected by the previously described procedure are separately re-processed. The IGT method is applied to the corresponding area of the original image, which is for a given area. The iterations stop when either the criterion of formula (2) is satisfied or the number of iterations exceeds the corresponding number of iterations required for the global thresholding on the entire image (from the first step of the proposed approach).

IGT removes a lot of pixels during the first iterations, given that the selected areas have relatively high average density of black pixels. The background noise in the selected areas is more likely to be removed since the area is likely to be more homogeneous than the entire image, in comparison to the application of IGT to the entire page. Generally, this procedure tends to move more pixels of the selected areas to the background in comparison with the previous application of IGT to the entire image. The foreground is not attenuated unless it consists of different grey-scale tones (e.g., presence of both printed and handwritten in the same area), if in case a selected area does not contain considerable amount of background noise.

An important factor for the successful re-application of IGT in selected areas is the size of the window nxn used in the procedure of selecting the appropriate areas (described in the previous subsection). A small window size forms more but smaller areas, which can be seen. There is advantage of adapting area in more detail at the part of the image that still has noise. In many cases they do not provide enough information for successfully re-apply the IGT algorithm on them and also the resulting areas are small. Fewer but bigger areas are detected, in case of large window size.

In order to effectively remove background noise, they provide enough information to the IGT algorithm. These areas cannot be easily adapted to a specific part of the image that still contains noise. The final image may contain neighboring areas that have dissimilar amount of background noise, as a consequence.

IV.RESULTS

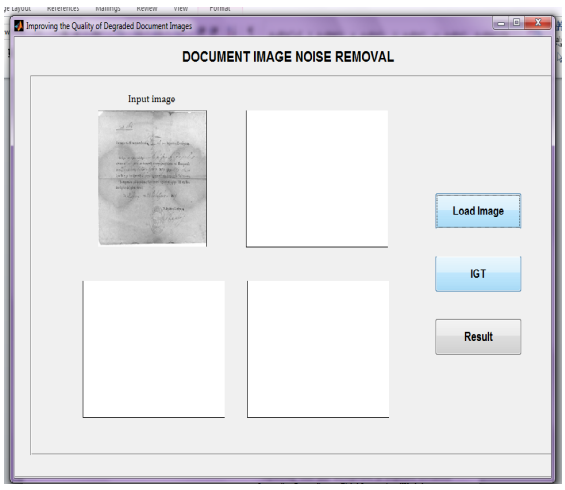


Fig 4.1 Window after loading the image from data set

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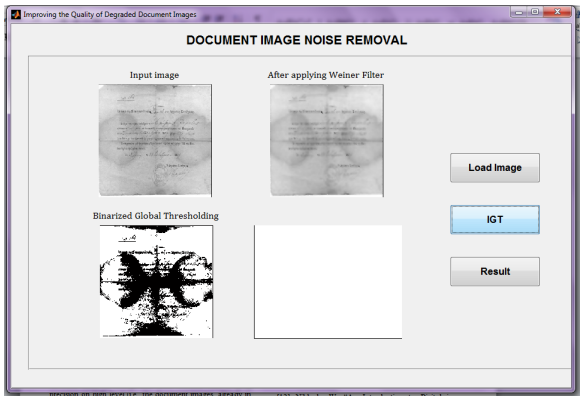


Fig 4.2 Window after preprocessing &IGT application

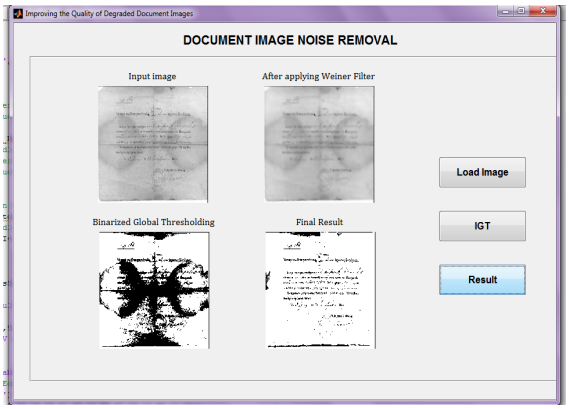


Fig 4.5 Window after enhancement of the old image

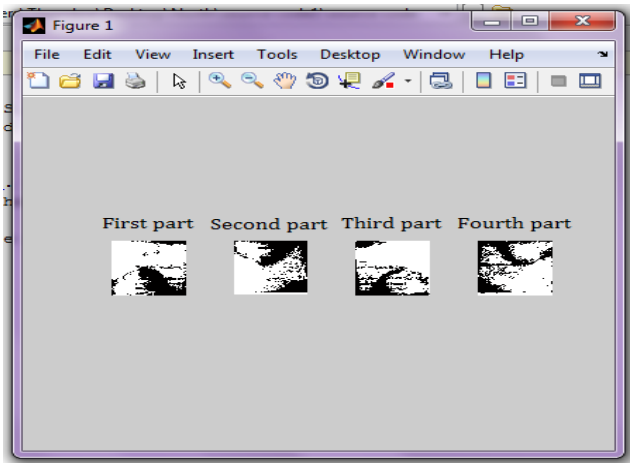


Fig 4.3 Window after local thresholding

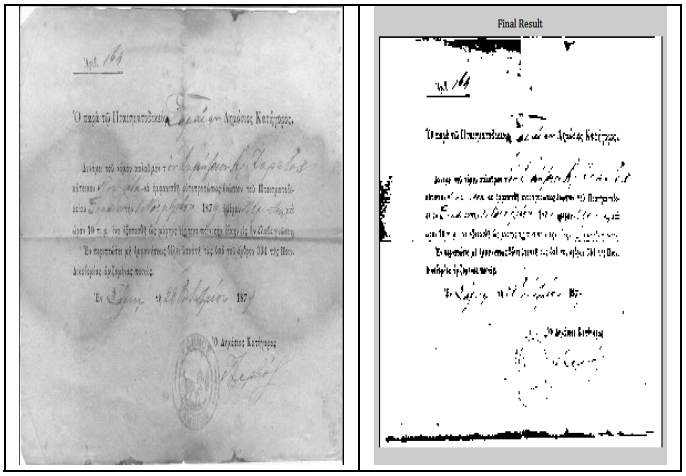


Fig 4.6 a)Original Image b)Enhanced Image

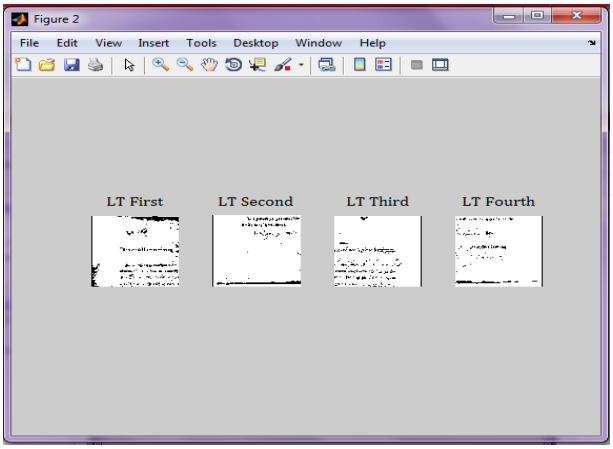


Fig 4.4 Window after local thresholding

V.CONCLUSION

In this paper a hybrid binarization approach is presented aiming at removal of background noise from the ancient and historical documents. An attempt of combining the advantages of local and global thresholding is made. The evaluation results using a historical document collection indicates that the proposed approach is able to deal with hard cases where different kinds of background noise coexist) while keeping precision on high level (i.e., the document images already in good condition are not affected). This process can be used in the framework of libraries who are willing to provide public access to their collection of historical documents.

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